



Torrefied biomass pellet production from rubber wood wastes and corn wastes in pilot-scale vibrating torrefaction system

Suneerat Fukuda^{*}, Jaggapan Sanduang, Supachita Krerkkaiwan

The Joint Graduate School of Energy and Environment (JGSEE) and Center of Excellence on Energy Technology and Environment, King Mongkut's University of Technology Thonburi, Bangkok 10140, Thailand

ARTICLE INFO

Keywords:

Biomass
Torrefaction
Pellet
Property improvement
Pilot scale
Cost analysis

ABSTRACT

Combining densification with torrefaction can substantially improve fuel properties, including energy content, grindability, and hydrophobicity, enabling torrefied pellets to replace coal in pulverized fuel boilers. This study investigates biomass pellet torrefaction using an innovative vibrating torrefaction system developed in this work, representing the first pilot-scale technical assessment of Thai biomass feedstocks. The objectives were to evaluate torrefied pellet properties and assess production costs. Rubber wood waste pellets (RWP) and corn waste pellets (CWP) were torrefied at 220–300 °C with residence times of 16 and 30 mins. The results indicate that increasing torrefaction severity enhanced fuel quality, with the heating value (HHV) reaching 22.4 MJ/kg, approximately 20% higher than raw pellets, along with improved grindability and moisture resistance. These improvements were accompanied by reductions in mass and energy yields, indicating that heating value alone is insufficient to assess process performance and that energy yield must also be considered. Specific electricity consumption ranged from 624 to 1161 kWh/ton of product, depending on torrefaction severity and mass yield. A preliminary cost analysis, accounting for capital investment, feedstock cost and electricity consumption, estimated production costs of 8000–11,000 THB/ton of torrefied pellets.

1. Introduction

As the world shifts toward cleaner and more sustainable energy systems, reducing reliance on coal has become a key priority due to its high greenhouse gas emissions and environmental impacts. Biomass has emerged as a promising substitute for coal in both industrial and power generation applications [1,2]. It is a renewable resource that can contribute to carbon neutrality when sourced and managed sustainably. In particular, the use of lignocellulosic biomass such as wood chips, agricultural residues, offers significant potential due to its vast availability, low cost, and compatibility with existing combustion technologies. However, the direct use of raw biomass is challenged by issues such as low energy density, high moisture content, and inconsistency in physical and chemical properties. These limitations can affect storage, transport, and combustion efficiency. To address these issues, pre-treatment technologies like torrefaction and pelletization are used to enhance fuel quality and make biomass more suitable as a coal replacement.

Torrefaction is a biomass pre-treatment process carried out at temperatures between 200 and 300 °C for 10 min to 2 h, typically in the

absence or presence of only low concentrations of oxygen [3]. This process generally yields around 70% of the original mass from woody biomass while retaining approximately 90% of its initial energy content [4]. Torrefaction modifies the chemical structure of biomass, increasing its energy density and reducing undesirable components such as moisture, volatile matter and oxygen. The optimal torrefaction conditions vary depending on the specific characteristics of the biomass feedstock. When processed under appropriate conditions, torrefied biomass can achieve fuel properties similar to those of low-rank coal [5,6], making it a promising alternative for coal substitution. Key industrial requirements for torrefied biomass include i) Water resistance – ability to resist degradation from moisture and humidity, ii) Grindability – ease of grinding similar to coal, and iii) High calorific value – improved energy content [7–9].

Despite those advantages, a key limitation remains. Torrefied biomass has relatively low bulk density, which poses challenges for storage, handling, and transportation. To address this, torrefaction is often combined with pelletization to produce torrefied pellets, also known commercially as black pellets. Torrefied pellets not only retain the improved fuel properties of torrefied biomass but also offer higher

^{*} Corresponding author.

E-mail address: suneerat.pip@kmutt.ac.th (S. Fukuda).

<https://doi.org/10.1016/j.fuproc.2026.108452>

Received 30 December 2025; Received in revised form 20 March 2026; Accepted 8 April 2026

Available online 19 April 2026

0378-3820/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

energy density and better physical durability. These enhancements make them more suitable for long-distance transport and industrial-scale use.

The production of torrefied pellets can follow two main process routes: torrefaction before pelletization (TBP) and torrefaction after pelletization (TAP) [10]. Each approach offers distinct advantages and challenges, depending on the desired fuel characteristics and processing requirements. The TBP approach allows for greater control over torrefaction conditions, making it easier to tailor the process to specific feedstocks. However, the torrefied biomass produced in this method is often brittle and more difficult to densify. This can result in lower pellet durability and increased energy consumption during pelletization. In contrast, the TAP method typically produces pellets with higher mechanical strength and bulk density, as the pellet structure remains intact during thermal treatment. It also simplifies material handling and allows for the simultaneous production of both conventional and torrefied pellets. Nevertheless, TAP may lead to less uniform torrefaction due to heat transfer limitations within the pellet core, and also requires precise control of temperature and residence time to achieve consistent product quality.

A variety of torrefaction technologies have been developed globally, ranging from laboratory-scale setups to full commercial-scale systems [11]. Some technologies are designed specifically for small particle sizes, such as sawdust, while others are capable of handling a broader range of biomass sizes [12–14]. These systems vary in complexity. Some are relatively simple to operate, while others require significant technical expertise and process knowledge. Many torrefaction technologies have been adapted from existing thermal processes. Among them, rotary drum or rotary kiln furnaces are the most widely developed and commercialized [15]. These systems, originally designed for drying applications, are valued for their mechanical simplicity and robust operation. However, they require significantly larger physical footprint than other technologies offering equivalent throughput. Additionally, they rely on combustion-based heating, which can make precise temperature control challenging, especially when managing large-volume reactors [16]. A promising alternative that addresses these limitations is the use of electric heating combined with vibration-assisted transport. With vibration, particle mixing is enhanced by its plug flow behavior. By controlling the vibration frequency, the residence time of all particles inside the reactor can be reasonably controlled compared to other reactors where there is a large residence span of the biomass particles [15,17]. Temperature inside the reactor can also be more precisely controlled by electricity as heating medium [18]. With these advantages, all particles would be properly and homogeneously torrefied before leaving the reactor. However, the properties of torrefied products and energy consumption must be investigated.

The studies using the vibration-assisted torrefaction are relatively limited, with most investigations conducted at the laboratory scale [19,20]. One notable exception is a pilot-scale torrefaction study by Doassans-Carrère et al. [18], which employed a 100 kg/h vibrating electrical elevator and reactor system. Originally developed for drying agricultural products in the food processing industry, their preliminary study showed that the reactor system was successfully adapted and validated for biomass torrefaction, demonstrating the flexibility of vibration-based systems in thermochemical applications.

This study investigates the torrefaction of biomass pellets using an innovative vibrating torrefaction technology. It represents the first technical assessment of Thai biomass feedstocks and the feasibility of producing torrefied biomass pellets at a pilot-scale level, utilizing a continuous vibrating torrefaction system developed during this research. The primary objective is to evaluate the characteristics of the torrefied pellets and analyze the associated production costs. Following the TAP route, biomass pellets derived from rubber wood waste (representing woody biomass) and corn waste (representing agricultural biomass) were selected as feedstocks for torrefaction, rather than using non-densified biomass. The impact of key operating parameters on the

yield and properties of the torrefied products was systematically examined. Additionally, the actual energy consumption was quantified to assess the energy demand, providing a foundation for evaluating the efficiency and performance of the torrefaction technology, as well as the overall production cost. Finally, the cost implications of torrefied pellet production using this technology are discussed.

2. Material and method

2.1. Biomass samples and analysis method

Two biomass feedstocks were used in this study including rubber wood pellet (RWP) and corn waste pellet (CWP). RWP was made from rubber wood wastes (mainly sawdust) from wood processing plants in Chonburi province where there are a lot of rubber tree plantation. Wood pellet industry already exists in Thailand for both export and domestic use and rubber wood waste is one of typical raw materials. Demand of torrefied wood pellets is also increasing worldwide. CWP was made from wastes after harvesting and processing corn grown in the highland area of Lampang province. Utilization of corn wastes is still limited and most of the wastes end up open burned, causing serious environmental problem in the northern part of Thailand. Business model to convert corn wastes into pellets and torrefied pellets will create a feasible market that encourages more collection of corn wastes.

Both raw materials were delivered from the local pellet plants in respective provinces to the test site in bulk. The pellets were later sampled and prepared for analysis of physical and chemical properties according to the methods and standards for testing biomass pellet fuels as listed in Table 1. The properties analyzed were based on the list of essential pellet parameters in the ENplus standard [21], which is one of the most widely acceptable wood pellet quality certification standards. Only the parameters related to the scope of this study were selected, including bulk density, pellet density, moisture content, proximate analysis (mainly for ash content), calorific value, and mechanical durability. In addition, grindability and suitability transportation, storage and handling were analyzed. Grindability was an indicator for energy requirement in grinding and hence suitability for applications like co-firing in power plants [22]. The pellet stability was assessed by the dissolved organic content in leachate, water absorption and equilibrium moisture uptake. The torrefied pellets were also subject to the same set of analysis for comparison.

Table 1
Analysis methods.

Property	Method
Bulk density	According to ASTM E 873–82 [23]
Pellet Density	Measurement and calculation of average ratio of weight per volume of 20 individual pellets
Moisture content	According to ASTM D 3173–11 [24]
Proximate analysis	Thermogravimetric Analysis [Shimadzu TA50] according to ASTM D7582–15 [25]
Ultimate analysis	CHN analyzer [J-Science Model JM-10] according to ASTM D5373–16 [26]
Calorific value	Bomb calorimetry according to ASTM E711–23e1 [27]
Mechanical Durability (DU)	According to ISO 17831–1:2025 [28]
Grindability	Hardgrove grindability index according to ASTM D409M–09 A [29]
Organic content in leachate (Chemical oxygen demand, COD)	Pellets were soaked in the purified water at 20 ± 1 °C for 72 h before the leachate was subject to analysis according to SM 2017 5220C [30]
Water absorption	Pellets were immersed in purified water at 20 ± 1 °C for 30 min. Water absorption was determined by the pellet weight before and after immersing. [31]
Equilibrium Moisture Content (EMC)	Pellets were oven-dried at 105 °C before placing in a water desiccator (100% relative humidity) at 20 ± 1 °C for 200 h. EMC was determined as the final weight of pellets after the test. [32]

2.2. Overview of torrefied biomass pellet production process and vibrating torrefaction reactor

Biomass in form of pellets were used as the raw material input for this torrefaction study. The process overview of torrefied pellet production from biomass following the torrefaction after pelletization (TAP) route is shown in Fig. 1. Biomass was ground to reduce particle size in the range of 0.4–3 mm to enhance the pelletizing process and ensuring consistent pellet quality. The ground biomass was then dried to reduce the moisture content to below 12%, which is suitable for pelletizing. Without any additives needed, lignin acted as a natural binder [33]. Through the industrial-scale pelletizer, the biomass particles were compressed and heated due to friction to form dense and uniform pellets with an approximately 8 mm in diameter. In this study, the biomass pellets were torrefied at 220–300 °C for 16 and 30 min. The obtained torrefied pellets are generally dark color, dry and brittle.

The torrefaction experiments were conducted using a vibrating torrefaction reactor, developed in this study as a pilot-scale prototype. The diagram of reactor, as illustrated in Fig. 2, consists of following main components.

- Raw material feeding system consisted of a screw feeder driven by electric motor and a fuel hopper. From the screw feeder calibration test, the amount of raw material entering the system could be varied by adjusting the inverter in the range of 0–30 kg/h. The calibration curves were different depending on the physical properties of the raw material.
- Vibrating reactor tower consisted of a tower and a stainless-steel spiral tube coiling up the tower. The tube was 100 mm in outer diameter and 50 m long. The diameter of each turn was about 3 m and the five turns covered the height about 2 m. During operation, the biomass particles would travel along in the tube by means of three-dimensional vibration created by the two off-balance vibrating motors equipped on the opposite sides of the tower. The residence time of pellet particles was varied at 16 and 30 min by adjusting the frequency of the vibration. A series of cold tests by using marked pellet particles were carried out to determine the residence time of pellet particles.
- Heating system was based on the concept of resistance heater by supplying the power to the steel tube, which then transferred heat to biomass. The heating mechanism is visualized as shown in Fig. 3(a). The main heat transfer mechanism was conduction, with less of convection from steam and hot gas generated by the reaction and radiation. The temperature at each zone (turn) was separately controlled by the power transformer (Heater no. 1–5) and the gas temperature inside the tube was monitored by the thermocouple (TC), denoted as T1–T5. The operation temperature was varied at 220, 250, 280 and 300 °C. While zone 2–5 were maintained the required torrefaction temperature, the temperature in zone 1 (feed drying zone) was set at 200 °C. The typical time-based temperature profiles at each zone in the spiral tube reactor (as indicated in Fig. 2) were presented in Fig. 3(b), which demonstrate the uniform and stable heat transfer enhanced by the vibration mechanism.
- Air injection/extraction system controlled the blower fan and exhaust fan through the inlet and outlet gas ducts installed at each coiling of the tube.

- Electrical control box

2.3. Operation procedure

The reactor was pre-heated by setting at the control system to the desired temperature for each zone (turn). When the desired temperatures were reached, the vibration motor was turned on at a specified frequency to control the residence time of biomass particles travelling inside the tube. The biomass was fed from the fuel hopper via the screw feeder at a specified feed rate. The inlet gas duct was kept closed while the outlet gas duct was open to release the product gases. The product steam and gases replaced the air inside the tube and were withdrawn by the exhaust fan through the stack. The biomass particles traveled up along the tube. The torrefied product exited at the top end and dropped into a water-cooled steel container (indirect cooling) to stop the reaction and cool down the torrefied product. The container was replaced with the new one every 5 min. The solid yield was determined by the ratio of product from direct weight measurement after cool down and the inlet feed of raw pellet. Representative samples from each condition were collected for property analyses. To end the experiment, the heating system and screw feeder were turned off, while the vibrating unit was kept on until no more biomass particles were left inside the tube. The exhaust fan was left on until no more product gases were observed. During the continuous operation, Actual electricity consumption to operate the torrefaction process was also measured by the build-in power meter recording, and calculated per unit weight of the feed and torrefied product.

2.4. Basic cost analysis

The basic cost analysis based on the test results from this study was carried out to reflect the key contributing cost components and how the operating conditions and torrefaction yields influenced those costs. The assumptions for cost analysis are listed as follows.

- All the costs were presented in Thai Baht (THB).
- Fixed cost covered only the cost of the pilot-scale 30 kg/h feed vibrating torrefaction reactor. The actual purchase cost was 3,500,000 THB. The expected lifetime was 10 years.
- Variable costs included only the electricity required for operating the torrefaction reactor and raw material pellets. The electricity tariff was 4 THB per kWh, which was an average value in the year 2023. The 6000 h of operation was assumed per year.
- The cost of RWP and CWP were 2400 and 2000 THB/ton. These values were the typical market price informed by the supplier of pellets used in this study. Since the operating cost (pelletizing) was similar, i.e. about 1000 THB/ton, this difference was primarily due to the cost of raw material. The raw material for RWP was the rubber wood wastes from wood processing in furniture factories, which can also be readily use or sold as fuel for boilers. CWP came from various residues of corn such as cobs, husks, stems and leaves, which are originally burned away at the field to clear the land. In average, the corn wastes were about 30–40% cheaper than rubber wood wastes.
- The cost of land and building, personnel cost and other utilities such as product cooling unit was not included.

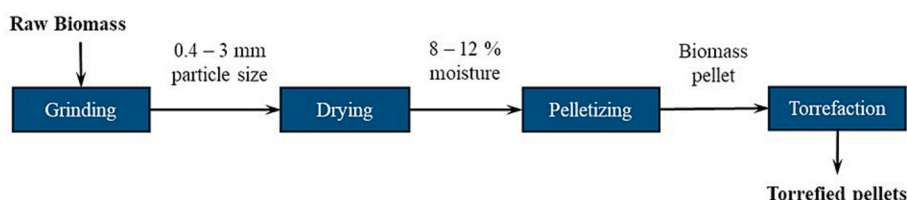


Fig. 1. The process overview of torrefied pellet production from biomass (TAP route).

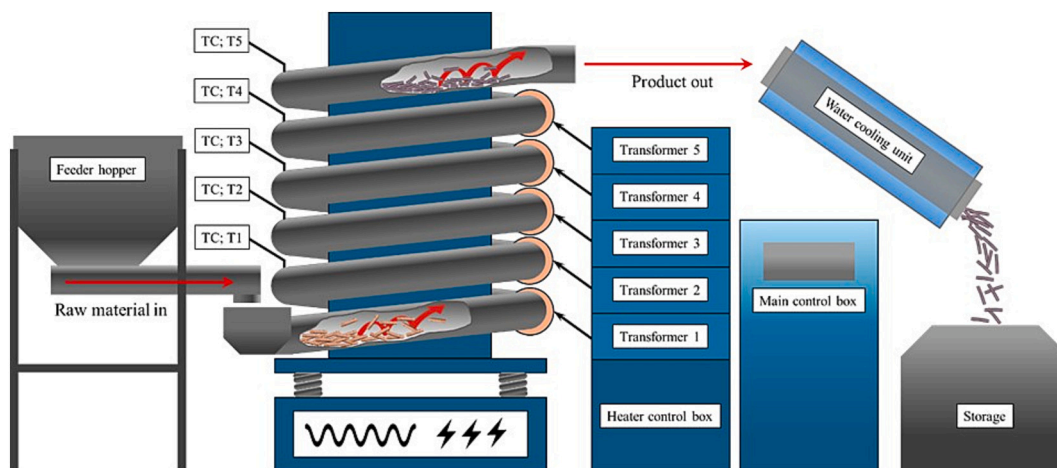
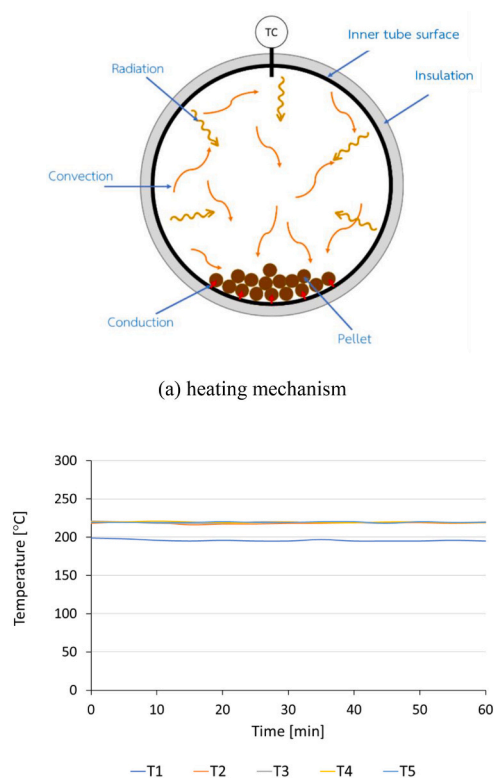


Fig. 2. Vibrating torrefaction system.



(b) Temperature profile at each zone for RWP torrefaction experiment at 220°C

Fig. 3. Heating system.

The total production cost per ton of torrefied product were then calculated based on the results of torrefaction yield and electricity consumption under each experimental condition. The results were also compared with the market prices or production cost reported in the literature.

3. Results & discussion

The experiments were carried out to investigate the effect of torrefaction temperature and residence time, which are important operational parameters determining the product yield and characteristics. The feed rates used for rubber wood pellet (RWP) and corn waste pellet (CWP) were 26.7 and 28.6 kg/h, respectively.

3.1. Effect of temperature and residence time on the torrefaction yield and colorization of torrefied product

The torrefaction or solid yields of RWP and CWP at different temperature and residence time are shown in Table 2. Both feedstocks similarly responded to the increase in the temperature and residence time. As the temperature increased from 220 to 300 °C, the solid yield of both feedstocks decreased significantly and constantly due to the decomposition of biomass leveraged with temperature. At 220 °C, around 12–14% mass loss was observed, while the mass loss became 40–45% at 300 °C. The low solid yield reflected the higher degree of thermal treatment and the consequent development of pellet color from brown (presenting the origin of biomass pellet) to black as shown in Fig. 4.

To investigate the scale effect on the system performance, the yield results of rubber wood pellets were compared with the lab-scale data of woody biomass pellets from relevant literature conducted at 225–300 °C [34–37]. Despite the differences in the scale of operation and biomass specificity, the results from this study did not significantly deviate from the yields observed from the lab-scale operation. This can be attributed to the superior heat and mass transfer which resulted from the advantageous features of the vibration torrefaction reactor, the combined effects of improved particle–wall contact and intensified mixing induced by vibration. As described in the system design, heat transfer in the reactor is primarily governed by conduction through the heated tube. The imposed vibration promotes continuous particle rearrangement and plug-flow behavior, reducing void spaces and increasing effective contact area between biomass pellets and the heated surface. This leads to a more uniform temperature distribution along the reactor and minimizes thermal gradients, as evidenced by the stable temperature profiles across reactor zones (Fig. 3(b)). At the microscale, vibration can also facilitate energy transfer across particle interfaces by reducing thermal resistance associated with poverty and agglomeration. Consequently, the system achieves relatively homogeneous torrefaction, contributing to consistent

Table 2

Effect of torrefaction temperature and residence time on the yield of torrefied RWP and CWP.

Temperature (°C)/Time (min)	Yield (% wt of solid)	
	RWP (26.7 kg/h)	CWP (28.6 kg/h)
220/16	86.7 ± 1.4	87.6 ± 0.5
220/30	82.7 ± 2.2	80.9 ± 1.5
250/16	77.4 ± 1.1	77.2 ± 1.1
280/16	63.9 ± 0.9	62.2 ± 2.3
300/16	58.2 ± 0.5	55.8 ± 2.2



(a) Rubber wood pellet (RWP)



(b) Corn waste pellet (CWP)

Fig. 4. Physical appearance of torrefied pellets prepared at different conditions compared to their respective raw pellets.

product quality across different operating conditions. However, to scale up further, the homogeneous heat and mass transfer through electrical heating and well-mixed solid transport must be maintained.

3.2. Effect of temperature and residence time on the torrefied product properties

The properties of the torrefied RWP and CWP from different conditions are shown in Tables 3 and 4, respectively. The properties of raw pellets are also provided as reference. It should be noted that a formal uncertainty analysis was not conducted for all the property analyses, as replicate measurements were not available for the datasets used in this study. While consistent trends are observed across the investigated conditions, the findings are considered indicative and should be confirmed by future studies with more comprehensive datasets.

3.2.1. Physical and chemical composition

Considering the properties of raw pellet, RWP had a low ash content (1.6%) and high volatile matter, which is the nature of woody biomass. On the other hand, CWP, which was made from agricultural waste materials, had high ash content (7.8%). This is one of the factors that resulted in a slightly lower heat content in CWP (17.68 MJ/kg) compared to RWP (18.6 MJ/kg).

During the torrefaction process, hemicellulose and partial cellulose decomposition occurs. Biomass loses the moisture and maintained the very low or almost no moisture content due to the hydrophobic property of torrefied product, which enables a long-term storage [38]. The release of moisture and volatiles resulted in the higher fixed carbon content and higher ash content in the torrefied product. At the maximum temperature tested (300 °C), the ash content of torrefied RWP is 3.7%, which is still relatively low, while that of torrefied CWP increased to 10.6%. The increased ash content may affect the combustion and ash handling for downstream application of the torrefied biomass [39]. Therefore, agricultural wastes (originally high ash content) may be less attractive feedstocks for torrefaction. Excessive accumulation of ash can also act as

thermal barriers while increase the total porosity of biomass [40].

The release of moisture and volatiles also resulted in an increase of C content and a decrease of H and O content. Considering the H/C and O/C values (calculated from the CHO results in Tables 3 and 4), as shown in Fig. 5, it was found that when the torrefaction temperature rises, the H/C and O/C ratio of the resulting product came closer to those of coal and charcoal.

3.2.2. Thermal property

The product heating value (MJ/kg) is an indicator determining the product quality and is often the requirement of subsequent energy conversion process such as co-processing with coal. Another indicator which should also be considered for the optimization of the torrefaction process is the energy yield (%), i.e. product yield (%) x heating value (MJ/kg). It reflects the energy conversion efficiency or, in other words, how much energy is lost during torrefaction process.

Fig. 6 presents the product heating value and energy yield for RWP and CWP. When the temperature was increased from 220 to 300 °C, the product heating value constantly increased by about 20% compared to the raw biomass. The heating value of torrefied RWP and CWP at 300 °C (which was the highest temperature studied) were 22.4 and 21.4 MJ/kg, respectively, comparable to that of lignite. However, the energy yield of both RWP and CWP decreased significantly due to the excessive mass loss at higher temperature. At 300 °C, the energy yield was around 70% and 67.4%, while the mass yield was 58.2 and 55.8% for RWP and CWP, respectively. Therefore, the energy yield should also be considered together with the product heating value for the selection of appropriate conditions in order to maintain the high efficiency of the process and its feasibility. Although increasing the residence time from 16 to 30 min only slightly increased heating value of the product, a large decrease in energy yield was attributed to a significant decrease of mass yield. When comparing the energy yield at the same torrefaction temperature between RWP (wood biomass) and CWP (non-wood biomass), it was found that the latter was generally lower. Non-wood biomass has a lower residual weight than wood biomass due to its higher content of hemicellulose (which can decompose at low temperatures). The results also agree with the literature that reported the energy yield of woody biomass in the range of 62.0–97.9%, while that of non-woody biomass such as grass or agricultural waste materials was wider at 26.9–94.6% [8,41].

3.2.3. Mechanical property

Pellet durability is a critical mechanical property that reflects the ability of pellets to withstand degradation from shocks and friction during transport, storage, and handling. The effectiveness of densification and pre-pelletting factors, such as feedstock characteristics affect the pellet durability [42], and consequently the behavior during torrefaction. In commercial practice, certain durability level is requested to ensure the minimum segregation during transportation, storage and at other stages of the production process. European standards require a minimum mechanical durability of 97.5% for premium quality pellets, while the PFI standard sets this threshold at 96.5% [43]. Better properties may be required for torrefied pellet production, since the product after torrefaction will have even lower density and become more fragile.

Both RWP and CWP had relatively good physical properties, i.e. high bulk density and %DU reaching that of premium quality pellets. Although with the decrease in bulk density and % DU, torrefied RWP still maintained its acceptable quality for transportation and long-term storage, especially %DU at 96.6% for torrefied RWP at 300 °C. On the other hand, the quality of torrefied CWP especially at high degree of torrefaction (i.e. 250 °C or above) is too poor with % DU below 90%. However, the mechanical durability of torrefied pellets from this study was in the high range, compared to other pilot-scale studies using various other torrefaction technologies as comparatively summarized in the study of Agar et al. [44].

Pulverized coal combustion boiler is the most widespread coal-fired

Table 3

The effect of temperature and residence time on the properties of torrefied rubber wood waste pellet (RWP) under different conditions.

Parameter	Raw RWP	Torrefaction condition (Temperature/Time)				
		220/16	220/30	250/16	280/16	300/16
<i>Physical and chemical composition</i>						
Moisture (wt%)	5.7	0.9	0.5	0.5	1.3	0.0
<i>Proximate Analysis (wt%, dry basis)</i>						
Volatile matter	82.9	81.9	81.5	80.7	73.7	65.8
Fixed carbon	15.4	15.5	16.4	16.9	24.1	30.5
Ash	1.6	2.5	2.1	2.4	2.2	3.7
<i>Ultimate analysis (wt%, dry basis)</i>						
C	46.7	48.0	47.6	48.4	52.8	55.6
H	7.1	6.6	6.8	6.5	6.2	6.2
N	0.4	0.4	0.4	0.5	0.5	0.5
O (by difference)	44.2	42.5	43.0	38.3	38.3	34.0
<i>Thermal property</i>						
Heating value, HHV (MJ/kg)	18.6	19.0	19.2	19.4	20.9	22.4
<i>Mechanical property</i>						
Bulk density (kg/m ³)	691.7	743.8	718.9	698.1	617.1	558.9
Pellet density (kg/m ³)	1230	1205	1201	1171	1050	978
Durability (%DU)	99.1	98.9	98.9	98.4	97.4	96.6
Hardgrove grindability index (HGI, –)	10	14	20	16	26	36
<i>Pellet stability</i>						
Organic matter in leachate of biomass pellets (COD, mg/L)	4618	4127	4133	4667	2720	1380
Water absorption (g H ₂ O/g dry matter)	0.44	0.15	0.13	0.11	0.10	0.07
Equilibrium Moisture Content (EMC), %	15.97	14.64	14.72	13.29	11.23	10.48

power generation technology in the world and in Thailand. When using biomass to replace coal, grindability is an important property that indicates the energy consumption in pulverization. This will result in the ability to increase the ratio of biomass in co-combustion with coal in

existing pulverized coal combustion boilers. Product grindability can be determined by several ways such as the particle size distributions after grinding [18], direct measurement of power consumed for grinding [9] and Hardgrove Grindability Index (HGI) [45]. This study used the HGI, a

Table 4

The effect of temperature and residence time on the properties of torrefied corn waste pellet (CWP) under different conditions.

Parameter	Raw CWP	Torrefaction condition (Temperature/Time)				
		220/16	220/30	250/16	280/16	300/16
<i>Physical and chemical composition</i>						
Moisture (wt%)	8.25	0.85	1.47	0.43	0.43	0.19
<i>Proximate analysis (wt%, dry basis)</i>						
Volatile matter	74.6	70.8	71.3	68.1	60.4	54.6
Fixed carbon	18.1	17.9	17.6	21.4	29.1	34.9
Ash	7.3	11.4	11.1	10.5	10.5	10.6
<i>Ultimate analysis (wt%, dry basis)</i>						
C	44.1	43.0	44.5	48.2	51.5	54.2
H	6.5	6.0	6.1	6.2	5.9	6.0
N	0.5	0.8	0.7	0.7	0.7	0.8
O (by difference)	41.5	38.9	37.6	34.5	31.4	28.5
<i>Thermal property</i>						
Heating value, HHV (MJ/kg)	17.7	17.3	17.5	18.8	20.2	21.4
<i>Mechanical property</i>						
Bulk density, kg/m ³	558.3	631.5	629.4	581.8	523.8	507.2
Pellet density (kg/m ³)	1108	1059	1064	1016	882	820
Durability (%DU)	98.2	96.9	97.3	94.0	90.4	88.0
Hardgrove grindability index (HGI, –)	16	18	18	26	37	42
<i>Pellet stability</i>						
Organic matter in leachate of biomass pellets (COD, mg/L)	5120	2300	3040	2940	910	483
Water absorption (g H ₂ O/g dry matter)	2.72	0.59	0.19	0.12	0.11	0.10
Equilibrium Moisture Content (EMC), %	17.31	12.74	11.46	11.37	10.30	10.02

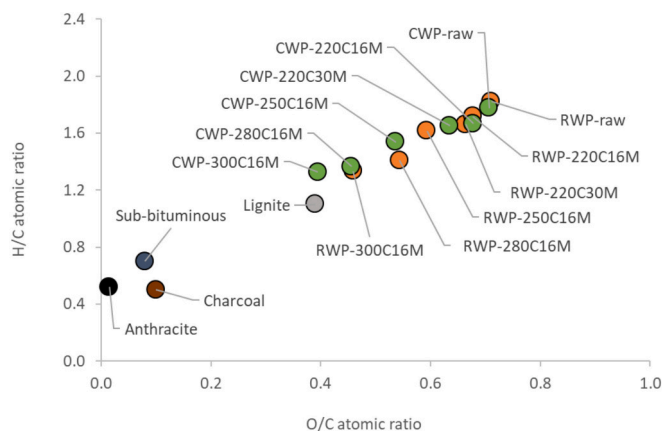


Fig. 5. H/C and O/C ratio of raw biomass and torrefied product prepared at different conditions vs. coal and charcoal (Data of coal and charcoal from Bridgeman et al. [5]).

measure for the grindability of coal, as the indicator for grindability. The higher HGI value means the easier grinding and therefore less energy consumption. The HGI value of RWP and CWP was found to increase with increased torrefaction temperature and residence time. The HGI values of pellets torrefied at 300 °C were 36 and 42 for RWP and CWP, respectively, or about 3 times higher than those of raw pellets. The values became comparable to those of lignite and low HGI sub-bituminous coals (i.e. HGI = 30–50 [45]), which enable co-milling in standard coal pulverizers. The HGI values of CWP were generally higher than those of RWP, for both raw and torrefied samples at the same temperature.

3.2.4. Pellet stability

Hydrophobicity or moisture resistance is an important indicator since moisture can have a critical effect on the physical stability and thermal property of pellets [46]. The hygroscopicity of the pellets should be as low as possible during the storage and transportation processes.

During storage, decomposition is concerned especially when exposed to moisture or rain. The hydrophobicity of torrefied biomass is known to prevent or slow down such reactions [38]. In this study, a leaching test was conducted by soaking pellets in water for 72 h at room temperature, after which the leachate was sent to COD analysis. During leaching, raw pellets and pellets torrefied at low temperature, i.e. 220 °C, exhibited swelling and segregation while the more severely torrefied products did not. By visual observation, the color of the leachate from torrefied pellets was clearer compared to the leachate from raw biomass and as the torrefaction temperature was increased. The results of COD analysis of the leachate, as shown in Tables 3 and 4, showed that when the temperature increased, the COD value of the leachate decreased, which suggests that the structure of the torrefied product became more stable and insoluble in water as the degree of torrefaction increased.

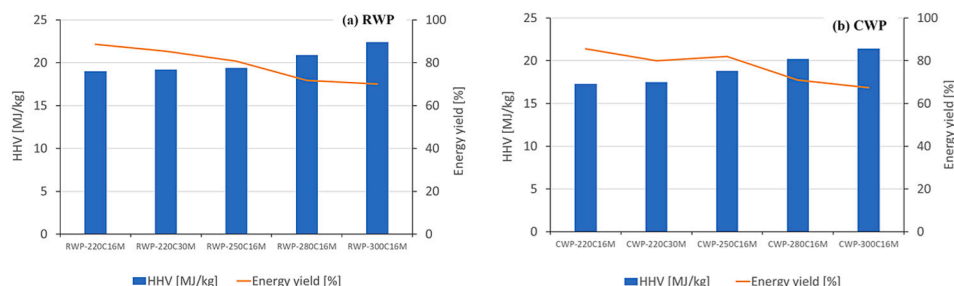


Fig. 6. Heating value vs. Energy yield of torrefied RWP and CWP.

Table 5

Electricity consumption rate at various operating conditions.

Sample name	Electricity consumption [kWh/ton]		Source
	Feed	Product	
RWP-220C16M	580	669	This study
RWP-220C30M	551	666	
RWP-250C16M	666	860	
RWP-280C16M	660	1033	
RWP-300C16M	675	1161	
CWP-220C16M	546	624	This study
CWP-220C30M	515	636	
CWP-250C16M	630	816	
CWP-280C16M	588	946	
CWP-300C16M	599	1073	
Torrefaction using pilot scale (50 kg/h) moving bed reactor	550		Nanou et al. [48]
Torrefaction in a designed full-scale operation (31.5 ton/h)	137		
Torrefaction using a larger pilot-scale (575 kg/h) resistance heater screw reactor	70–190		Severy et al. [49]

The results of water absorption and equilibrium moisture content (EMC) also confirmed the increased hydrophobicity of pellets by torrefaction treatment. Water absorption of RWP pellets (0.44 g H₂O/g dry matter) drastically dropped even for pellets torrefied at 220 °C for 16 min (0.15 g H₂O/g dry matter) and further reduced at higher degree of torrefaction. Similar trend was observed for CWP. This coincided with the reduction in EMC compared to raw pellets. Under the conditions studied (from the lowest severity (220 °C) to highest severity (300 °C)), the reduction in EMC was increased from 8% to 34% for torrefied RWP and from 26% to 42% for torrefied CWP. However, as the degree of torrefaction reached the highest, the hydrophobicity property of torrefied pellets from both biomass origin became similar. Small cracks or changes in pellet form were noticed for pellets torrefied at 280 °C or above.

It is clear that in term of pellet stability, torrefaction at above 250 °C is recommended. These results would be beneficial for long term storage of pellets, especially in open areas when pellets are exposed to humid environmental conditions. The better water resistance property in torrefied pellets helps inhibit the biological degradation and spoilage [47].

3.3. Energy consumption for torrefaction process

In addition to fuel cost, energy cost is also a key factor for torrefied biomass production cost. The source of energy to operate the vibrating torrefaction reactor developed in this study was electricity, which was used mainly for heating. The electricity was consumed for heating the system (the reactor tube), heating the incoming biomass feedstocks and wasted as energy losses. The electricity consumption was measured and calculated per unit weight of the feed and torrefied product.

The results of specific electricity consumption under various operating conditions are shown in Table 5. It was found that the electricity

consumption for the production of torrefied pellet was in the range of 515–675 kWh/ton feed or 624–1161 kWh/ton product (depending on degree of torrefaction and mass yield of each condition). Electricity consumption tended to increase when higher torrefaction temperatures were required. The increase was more pronounced when estimated for kWh/ton product. For the same temperature and residence time, producing torrefied CWP required lower electricity consumption than torrefied RWP. One reason may be due to the feed rate of CWP (28.6 kg/h) was higher than that of RWP (26.7 kg/h) and therefore lower electricity consumption when normalized per unit of product.

Electricity use analysis was conducted by comparing the electricity use for heating the system with and without feedstocks under otherwise similar temperature and vibrating conditions. It was found that about 2/3 of the total electricity supplied to the system was used to maintain the temperature of reactor tube. The energy transferred to heat the biomass was only 1/3, or about 200 kWh/ton feed, which is in line with theoretical calculation reported by Doassans-Carrère et al. [18] that energy used in the torrefaction process at temperatures of 230–250 °C was in average 770 kJ/kg or 215 kWh/ton feed. However, there was no report of actual measurement. The finding implies that there was a reasonable amount of heat loss in the system.

The specific electricity consumption of the vibrating torrefaction system developed in this study was also compared with other torrefaction systems. Despite the limited information in the literature and the difficulty to make a comparison of reported efficiencies which may be greatly affected by factors such as technology, capacity, fuel properties (especially moisture), the following studies were comparable to this study. Nanou et al. [48], which studied torrefaction using pilot scale (50 kg/h) moving bed reactor, reported that the energy consumption for torrefaction was about 2 GJ/ton of feedstocks or equivalent to 550 kWh/ton of feedstocks. The data obtained from the pilot scale experiments were used for the design of the full-scale operation (31.5 ton/h). The energy consumption was reduced to 137 kWh/ton of feedstocks. Severy et al. [49] studied torrefaction using a larger pilot-scale (575 kg/h) resistance heater screw reactor. The energy consumption varied at 70–190 kWh/ton of raw materials depending on the temperature used, the humidity of the raw materials and the feed rate. From the above references, it is clear that the capacity of the system has a significant impact on the specific energy consumption. System scale-up would greatly reduce the energy losses.

3.4. Basic cost analysis

The estimated production costs of torrefied pellets based on the

Table 6
Production cost and cost distribution of torrefied pellets.

Condition	Cost (THB/ton torrefied product)			Source
	Fixed cost	Raw material cost	Energy cost	
RWP 250	2822.70	3100.78	3441.86	This study
RWP 280	3419.04	3755.87	4131.46	
CWP 250	2642.00	2590.67	3264.25	
CWP 280	3279.14	3215.43	3781.35	
2025 FOB price of black pellets from Thailand	6500–8000 THB/ton			This study
Minimum selling price (MSP) of torrefied wood pellet	\$211/ton			Agar [50]
Minimum selling price (MSP) of torrefied wood pellet	\$207/ton for the Torrefaction before Pelletization			Manouchehrinejad et al. [46]
	\$197/ton for Torrefaction After Pelletization			

assumptions in section 2.4 are presented in Table 6, comparing rubberwood pellets (RWP) and corn waste pellets (CWP) torrefied at 250 °C and 280 °C. It is important to note again that the production costs presented here covered only some key cost components that are the cost of torrefaction reactor (fixed cost), pellets (raw material cost) and electricity consumed for torrefaction process (energy cost). Among them, the cost of energy consumption had the most significant impact on overall production costs. Higher torrefaction temperatures not only reduced product yield but also increased the energy usage, resulting in a higher cost per unit of product. Specifically, torrefied pellets produced at 280 °C were approximately 20% more expensive than those produced at 250 °C. In terms of feedstock, torrefied CWP was about 10% less costly than torrefied RWP, primarily due to the lower cost of raw corn waste pellets.

Although the market price of torrefied pellets varies depending on calorific value and other performance characteristics, an information from a local black pellet producer revealed that the 2025 FOB price of black pellets from Thailand ranged from 6500 to 8000 THB/ton, depending on the degree of torrefaction and the purchased price agreement. Compared to such price range, the production cost based on this study, which ranged between 8000 and 11,000 THB/ton, was still too high, especially for pellets with high degree of torrefaction. Further improvement to lower the costs is needed. Nonetheless, torrefaction at temperature of 250 °C seemed to offer a more favorable trade-off between energy enhancement, product yield, product physical properties, and economic viability. Torrefaction at lower temperature is not recommended, due to the resulting low heating values and poor stability.

The production cost data from this study were also compared with values in other studies from the literature. Agar [50] reported the minimum selling price (MSP) of torrefied wood pellet at \$211/ton based on 2017 USD. Manouchehrinejad et al. [51] presented that, based on 100,000 ton of plant capacity per year, the MSP of torrefied wood pellet was \$207/ton and \$197/ton for the Torrefaction before Pelletization and Torrefaction After Pelletization configurations, respectively. Since the properties of torrefied products were not specified but would have significant influence on the price, it is difficult to compare on the same basis. However, the reported values in the literature were on the lower side of the values found from this study.

The basic cost analysis above underscores the importance of process optimization in enhancing the economic viability of torrefied pellet production. Costs can be further reduced by sourcing lower-cost raw materials and accessing cheaper electricity. Furthermore, reducing specific electricity consumption, either through improved process energy efficiency, scaling up production, or both, can significantly strengthen the financial sustainability of torrefied pellet manufacturing. Some possible solutions to improve energy efficiency include the energy recovery of torgas as partial heat source, the heat transfer enhancement by increasing the contact area between the tube wall and the pellet particles. Our initial tests by introducing the hot inert gas into the system exhibited a positive trend in reducing the electricity consumption of the system. However, a complete mass and energy balance are needed for the improved system design.

3.5. Analysis and implication for industrial-scale implementation

Globally, the transition of torrefaction technology to full industrial deployment remains limited and technically challenging. Several commercial-scale plants, such as the Topell facility in the Netherlands [52], have demonstrated production capacities ranging from tens of thousands to over 100,000 t yr⁻¹, confirming the feasibility of torrefied biomass (black pellets) as a coal substitute. However, early industrial implementations have revealed persistent operational issues, including reactor fouling, difficulties in heat integration, feedstock heterogeneity, and high energy consumption. These factors have contributed to inconsistent product quality and reduced overall process efficiency, highlighting the complexity of scaling torrefaction systems.

The results of this study demonstrated the “strengths” of the pilot-scale torrefaction in the production of biomass pellets with enhanced fuel properties, higher heating value, improved grindability, and increased hydrophobicity, making them suitable for direct co-firing in existing coal-based systems while utilizing locally available wood and agricultural residues. The vibration technology has advantage in the process intensification, enabling near plug-flow behavior and minimizing residence time dispersion, an inherent limitation in many conventional reactors. This results in more uniform devolatilization and consistent product quality, which is critical for downstream applications such as co-firing or intermediates for chemical syntheses. The electrically heated reactor allows precise temperature control within the torrefaction range of 220–300 °C, which is essential for controlling the balance between fuel quality improvement and mass yield loss. The system may therefore be well suited for small-to-medium scale decentralized biomass processing facilities, or integrated pellet production plants where torrefaction is applied after pelletization (TAP).

The findings also provided valuable operational data for industrial scale-up, particularly in terms of energy consumption and process efficiency. However, this also reflects the “weaknesses”. The system is fundamentally constrained by high specific energy consumption. The reliance on electricity as the primary heat source further introduces economic and carbon intensity concerns, particularly in regions with fossil-based grids. The vibrating transport mechanism is most effective for relatively uniform and mechanically stable particles such as pellets, chips or grains; irregular raw biomass particles with wide size distribution and in fibrous form may not move uniformly inside the spiral tube and could lead to inconsistent residence times or potential blockages. Maintaining pellet durability under severe torrefaction conditions is also crucial and has to be ensured by the sufficiently high density of raw pellet. Additionally, the mechanical complexity associated with vibration systems raises concerns regarding operational reliability, maintenance costs, and long-term durability, especially under continuous industrial conditions.

Significant “opportunities” exist for system-level optimization and integration, particularly through the recovery and utilization of torrefaction gases (torgas) as an internal heat source, which could substantially reduce external energy demand. Liquid or condensable fraction can be a source for value added chemicals, e.g. 5-HMF, at a suitable condition [53]. Since the collection and characterization of the liquid product were not carried out in this study, these have been recommended for future studies to understand the mass and energy balance as well as environmental implications of the process.

Scaling up the system offers potential for improved thermal efficiency through reduced surface-area-to-volume ratios, addressing one of the key weaknesses observed at pilot scale. The technology is well positioned to valorize low-cost and underutilized biomass residues, especially in regions with abundant agricultural wastes. Under increasing global pressure for coal phase-out and negative-emission energy systems, turning these wastes into pellets and torrefied pellets to help decarbonize power and industry sectors enhances their strategic relevance, particularly if integrated with carbon capture. This approach also benefits the society as a whole by offering a pathway alternative to open-field burning of agricultural wastes lowering emissions of greenhouse gases and environmental hazardous PM_{2.5}, and strengthen sustainable bioenergy value chains.

However, these opportunities are counterbalanced by several “threats”. The technology has strong competition not only from low-cost fossil fuels, but also conventional biomass pellets, and thermally self-sustaining torrefaction systems (e.g., rotary kilns and moving beds) that can utilize autothermal operation, thereby achieving significantly lower operational costs. Without substantial improvements in energy efficiency, the vibrating system risks being economically uncompetitive at commercial scale. Moreover, electricity price volatility and grid carbon intensity pose critical risks to both cost stability and environmental performance. Feedstock heterogeneity remains a persistent challenge,

potentially amplifying process variability despite improved reactor control, while feedstock supply stability determines the feasibility and sustainability of the project. Finally, uncertainties in policy support, carbon pricing mechanisms, and market standardization for torrefied biomass may hinder large-scale adoption, particularly in emerging markets where cost sensitivity is high.

Forward looking into wider and higher technology level application is worth exploring to improve the overall economic viability and resource efficiency. Apart from introducing torrefied biomass for direct co-firing in existing coal boilers at higher blend ratios compared to raw biomass [39,42], it can be applied to other downstream utilizations. Torrefied biomass was reported to improve overall power production efficiency of the downstream gasification process [54], to replace pulverized coal (PCI) and metallurgical coke in the blast furnace, enhancing low-carbon iron and steel production [55], and improve bio-oil quality during catalytic pyrolysis [56]. Catalytic pyrolysis is particularly interesting under the biorefinery concept, where lignocellulosic biomass is converted into multiple value-added products. Through cascading conversion routes, bio-oil can be upgraded into transportation fuels, syngas can be used for power or hydrogen production, and biochar can serve as a soil amendment, carbon sink, or catalyst precursor.

4. Conclusion

This study conducted the technical assessment and basic cost analysis of torrefied pellet production from pellets of two biomass feedstocks (rubber wood waste and corn waste) using a pilot-scale vibrating torrefaction system developed in this study. Product characteristics according to the international pellet standard and typical properties of torrefied biomass for coal replacement were comprehensively investigated. Increasing torrefaction temperature (i.e. in the range of 220–300 °C) and residence time of pellets inside the reactor (i.e. 16 and 30 min) improved both physical and thermal properties of pellets, closer to those of coal. Considering the heat content and energy yield of torrefied product, 250 °C seems to be a compromise, above which the heat content (HHV) sharply increased but the energy yield dropped considerably due to low torrefaction yield. The basic production costs were estimated 8000–11,000 THB/ton of torrefied pellet, depending on the feedstock type and torrefaction condition. Cost reduction is necessary and can be achieved through the use of lower-cost feedstocks and energy sources, along with improving energy efficiency and system scale-up.

CRediT authorship contribution statement

Suneerat Fukuda: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Jaggapan Sanduang:** Writing – original draft, Investigation, Formal analysis, Data curation. **Supachita Krerkkaiwan:** Writing – original draft, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Financial support from the Program Management Unit for Competitiveness (PMUC), Office of National Higher Education Science Research and Innovation Policy Council (NXPO) (Grant no. C10F630265) is gratefully acknowledged. The authors sincerely thank Weerachai Limpornchaicharoen for his technical advice and information on commercial aspects.

Data availability

Data will be made available on request.

References

- [1] F. Sher, A. Yaqoob, F. Saeed, S. Zhang, Z. Jahan, J.J. Klemes, Torrefied biomass fuels as a renewable alternative to coal in co-firing for power generation, *Energy* 209 (2020) 118444, <https://doi.org/10.1016/j.energy.2020.118444>.
- [2] N. Yadav, G. Yadav, V. Bakthavachalam, L. Potturaja, J.K. Roy, S. Elumalai, Agro-industrial residue torrefaction to bio-coal: its physico-chemical characterization and potential applications in energy and environmental protection, *Bioresour. Technol.* 418 (2025) 131948, <https://doi.org/10.1016/j.biortech.2024.131948>.
- [3] S. Konsomboon, J.-M. Commandre, S. Fukuda, Torrefaction of various biomass feedstocks and its impact on the reduction of tar produced during pyrolysis, *Energy Fuel* 33 (4) (2019) 3257–3266.
- [4] P.C.A. Bergman, A.R. Boersma, R.W.R. Zwart, J.H.A. Kiel, 2025. Torrefaction for Biomass Co-firing in Existing Coal-Fired Power Stations. Energy Research Centre of the Netherlands (ECN), 1–71 (Ecn05013).
- [5] T.G. Bridgeman, J.M. Jones, I. Shield, P.T. Williams, Torrefaction of reed canary grass, wheat straw and willow to enhance solid fuel qualities and combustion properties, *Fuel* 87 (2008) 844–856, <https://doi.org/10.1016/j.fuel.2007.05.041>.
- [6] B. Yan, W. Chen, X. Sun, X. Zhu, D. Hou, K. Yang, S. Zhou, G. Chen, Biomass torrefaction coupled gasification technology: a critical review, *J. Energy Inst.* 123 (2025) 102224, <https://doi.org/10.1016/j.joei.2025.102224>.
- [7] T.G. Bridgeman, J.M. Jones, A. Williams, D. Waldron, An investigation of the grindability of two torrefied energy crops, *Fuel* 89 (2010) 3911–3918, <https://doi.org/10.1016/j.fuel.2010.06.043>.
- [8] M. Phanphanich, S. Mani, Impact of torrefaction on the grindability and fuel characteristics of forest biomass, *Bioresour. Technol.* 102 (2011) 1246–1253, <https://doi.org/10.1016/j.biortech.2010.08.028>.
- [9] J.S. Tumuluru, S. Sokhansanj, J.R. Hess, C.T. Wright, R.D. Boardman, A review on biomass torrefaction process and product properties for energy applications, *Ind. Biotechnol.* 7 (2011) 384–401, <https://doi.org/10.1089/ind.2011.7.384>.
- [10] M. Manouchehrinejad, S. Mani, Torrefaction after pelletization (TAP): Analysis of torrefied pellet quality and co-products, *Biomass Bioenergy* 118 (2018) 93–104, <https://doi.org/10.1016/j.biombioe.2018.08.015>.
- [11] J. Koppejan, S. Sokhansanj, S. Melin, S. Madrali, Status Overview of Torrefaction Technologies. IEA Bioenergy Task 32 Report – Final Report, 2012.
- [12] Z.L. Wang, C.J. Lim, J.R. Grace, H. Li, M.R. Parise, Effects of temperature and particle size on biomass torrefaction in a slot-rectangular spouted bed reactor, *Bioresour. Technol.* 244 Part 1 (2017) 281–288, <https://doi.org/10.1016/j.biortech.2017.07.097>.
- [13] A. Javanmard, M.F.A. Patah, A. Zulhelmi, W.M.A.W. Daud, A comprehensive overview of the continuous torrefaction method: operational characteristics, applications, and challenges, *J. Energy Inst.* 108 (2023) 101199, <https://doi.org/10.1016/j.joei.2023.101199>.
- [14] S.L. Majama, T.A. Amibo, Modeling and optimization of chemical-treated torrefaction of wheat straw to improve energy density by response surface methodology, *Biomass Convers. Biorefinery* 14 (2024) 21213–21227, <https://doi.org/10.1007/s13399-023-04192-y>.
- [15] M. Creemers, J. Koppejan, J. Middelkamp, J. Witkamp, S. Sokhansanj, S. Melin, S. Madrali, Status overview of torrefaction technologies - A review of the commercialisation status of biomass torrefaction, IEA Bioenergy Task 32: Biomass Combustion and Cofiring (2015) 14–21.
- [16] S. Graham, J.M. Jones, M. Dekker, An integrated laboratory and industrial scale study of autothermal torrefaction of hardwood, softwood and miscanthus, *Biomass Bioenergy* 195 (2025) 107723, <https://doi.org/10.1016/j.biombioe.2025.107723>.
- [17] P. Piersa, H. Unyay, S. Szufa, W. Lewandowska, R. Modrzewski, R. Ślęzak, S. Ledakowicz, An extensive review and comparison of modern biomass torrefaction reactors vs. biomass pyrolysis—Part 1, 2022. *Energies* 15(6), 2227. doi:<https://doi.org/10.3390/en15062227>.
- [18] N. Doassans-Carrère, S. Muller, M. Mitzkat, REVE – a new industrial technology for biomass torrefaction: pilot studies, *Fuel Process. Technol.* 126 (2014) 155–162, <https://doi.org/10.1016/j.fuproc.2014.04.026>.
- [19] S. Sonsupap, Improvement of *leucaena leucocephala* Properties by Torrefaction in a Vibrating Reactor, M.Eng. Thesis, Mahasarakham University, 2018, <https://202.28.34.124/dspace/handle/123456789/195>.
- [20] P. Keeratiisariyakul, P. Rousset, A. Pattiya, Coupled effect of torrefaction and densification pre-treatment on biomass energetic and physical properties, *J. Sustain. Energy Environ.* 10 (2019) 97–106.
- [21] EN, ENplus® Wood Pellets – Requirements for Companies ENplus® ST 1001:2022, First edition, 2022. www.enplus-pellets.eu.
- [22] M.V. Gil, R. García, C. Pevida, F. Rubiera, Grindability and combustion behavior of coal and torrefied biomass blends, *Bioresour. Technol.* 191 (2015) 205–212, <https://doi.org/10.1016/j.biortech.2015.04.117>.
- [23] ASTM, ASTM E873–82(2019) Standard Test Method for Bulk Density of Densified Particulate Biomass Fuels, 2019.
- [24] ASTM, ASTM D3173–11 Standard Test Method for Moisture in the Analysis Sample of Coal and Coke, 2011.
- [25] ASTM, ASTM D7582–15 Standard Test Methods for Proximate Analysis of Coal and Coke by Macro Thermogravimetric Analysis, 2015.
- [26] ASTM, ASTM D5373–16 Standard Test Methods for Determination of Carbon, Hydrogen and Nitrogen in Analysis Samples of Coal and Carbon in Analysis Samples of Coal and Coke, 2016.
- [27] ASTM, ASTM E711–23e1 Standard Test Method for Gross Calorific Value of Refuse-Derived Fuel by the Bomb Calorimeter, 2023.
- [28] ISO, ISO 17831-1:2025 Solid Biofuels - Determination of Mechanical Durability of Pellets and Briquettes - Part 1: Pellets (ISO 17831-1:2025), 2025.
- [29] ASTM, ASTM D409/D409M-09a Standard Test Method for Grindability of Coal by the Hardgrove-Machine Method, 2009.
- [30] SM, SM 5220C: COD by Closed Reflux, Titration, 2017.
- [31] P. Matkowski, A. Lisowski, A. Świętochowski, Characterisation of wheat straw pellets individually and in combination with cassava starch or calcium carbonate under various compaction conditions: determination of pellet strength and water absorption capacity, *Materials* 13 (19) (2020) 4375, <https://doi.org/10.3390/ma13194375>.
- [32] I.D. Hartley, L.J. Wood, Hygroscopic properties of densified softwood pellets, *Biomass Bioenergy* 32 (2008) 90–93, <https://doi.org/10.1016/j.biombioe.2007.06.009>.
- [33] J. Anno-Kusi, T.R. Sarker, S. Nanda, A.K. Dalai, Pelletization of Torrefied Biomass using Binders, in: S. Nanda, V.D.N. Vo (Eds.), *Progressive Thermochemical Biorefining Technology*, 2021, pp. 105–124, <https://doi.org/10.1201/9781003098591>.
- [34] L. Wang, L. Riva, O. Skreiberg, R. Khalil, P. Bartocci, Q. Yang, H. Yang, X. Wang, D. Chen, M. Rudolfsson, H.K. Nielsen, Effect of torrefaction on properties of pellets produced from woody biomass, *Energy Fuel* 34 (2020) 15343–15354, <https://doi.org/10.1021/acs.energyfuels.0c02671>.
- [35] N. Setkit, X. Li, H. Yao, N. Worasuwanarak, Torrefaction behavior of hot-pressed pellets prepared from leucaena wood, *Bioresour. Technol.* 321 (2021) 124502, <https://doi.org/10.1016/j.biortech.2020.124502>.
- [36] P. Brachi, R. Chirone, M. Miccio, G. Ruoppolo, Fluidized bed torrefaction of commercial wood pellets: Process performance and solid product quality, *Energy Fuel* 32 (9) (2018) 9459–9469, <https://doi.org/10.1021/acs.energyfuels.8b01519>.
- [37] L.E. Hernandez-Mena, E. García, I. Fernandez, Y.L. Pereira, W.A. Bizzo, Torrefaction of wood pellets in Uruguay: effect in the energy properties, *Chem. Eng. Trans.* 109 (2024) 631–636, <https://doi.org/10.3303/CET24109106>.
- [38] W.-H. Chen, B.-J. Lin, Y.-Y. Lin, Y.-S. Chu, A.T. Ubando, P.L. Show, H.C. Ong, J.-S. Chang, S.-H. Ho, A.B. Culaba, A. Pétrissans, M. Pétrissans, Progress in biomass torrefaction: principles, applications and challenges, *Prog. Energy Combust. Sci.* 82 (2021) 100887, <https://doi.org/10.1016/j.pecs.2020.100887>.
- [39] S.V. Vassilev, C.G. Vassileva, Y.-C. Song, W.-Y. Li, J. Peng, Ash contents and ash-forming elements of biomass and their significance for solid biofuel combustion, *Fuel* 208 (2017) 377–409, <https://doi.org/10.1016/j.fuel.2017.07.036>.
- [40] S. Zhang, Y. Zhang, X. Zhang, D. Xing, J. Hu, Z. Zhao, L. Yao, Physicochemical properties of torrefied biomass at different atmosphere: a review, *Ind. Crop. Prod.* 236 (2025) 121988, <https://doi.org/10.1016/j.indcrop.2025.121988>.
- [41] Y. Uemura, W.N. Omar, T. Tsutsui, S.B. Yusup, Torrefaction of oil palm wastes, *Fuel* 90 (2011) 2585–5891, <https://doi.org/10.1016/j.fuel.2011.03.021>.
- [42] O. Williams, S. Taylor, E. Lester, S. Kingman, D. Giddings, C. Eastwick, Applicability of mechanical tests for biomass pellet characterisation for bioenergy applications, *Materials* 11 (2018) 1329, <https://doi.org/10.3390/ma11081329>.
- [43] J.S. Tumuluru, Effect of pellet die diameter on density and durability of pellets made from high moisture woody and herbaceous biomass, *Carbon Resour. Convers.* 1 (1) (2018) 44–54, <https://doi.org/10.1016/j.crcon.2018.06.002>.
- [44] D.A. Agar, M. Rudolfsson, S. Lavergne, T. Melkior, D.D.S. Perez, C. Dupont, M. Campargue, G. Kalen, S.H. Larsson, Pelletizing torrefied biomass at pilot-scale - quality and implications for co-firing, *Renew. Energy* 178 (2021) 766–774, <https://doi.org/10.1016/j.renene.2021.06.094>.
- [45] M.O. Obradović, D.B. Radić, D.M. Todorović, A.M. Jovović, N.V. Karličić, M. Stanojević, Practical assessment of grinding capacity and power consumption based on hardgrove grindability index and coal characteristics, *Therm. Sci.* 23 (5) (2019) 1533–1542, <https://doi.org/10.2298/TSCI1806053760>.
- [46] C. Whittaker, I. Shield, Factors affecting wood, energy grass and straw pellet durability - a review, *Renew. Sust. Energ. Rev.* 71 (2017) 1–11, <https://doi.org/10.1016/j.rser.2016.12.119>.
- [47] W.-H. Chen, J. Peng, X.T. Bi, A state-of-the-art review of biomass torrefaction, densification and applications, *Renew. Sust. Energ. Rev.* 44 (2015) 847–866, <https://doi.org/10.1016/j.rser.2014.12.039>.
- [48] P. Nanou, M.C. Carbo, J.H.A. Kiel, Detailed mapping of the mass and energy balance of a continuous biomass torrefaction plant, *Biomass Bioenergy* 89 (2016) 67–77, <https://doi.org/10.1016/j.biombioe.2016.02.012>.
- [49] M.A. Severy, C.E. Chamberlin, A.J. Eggink, A.E. Jacobson, Demonstration of a pilot-scale plant for biomass torrefaction and briquetting, *Appl. Eng. Agric.* 34 (1) (2018) 85–98, <https://doi.org/10.13031/AEA.12376>.
- [50] D.A. Agar, A comparative economic analysis of torrefied pellet production based on state-of-the-art pellets, *Biomass Bioenergy* 97 (2017) 155–161, <https://doi.org/10.1016/j.biombioe.2016.12.019>.
- [51] M. Manouchehrinejad, E.M.T. Bilek, S. Mani, Techno-economic analysis of integrated torrefaction and pelletization systems to produce torrefied wood pellets, *Renew. Energy* 178 (2021) 483–493, <https://doi.org/10.1016/j.renene.2021.06.064>.
- [52] <https://www.blackwood-technology.com/company/references/>.
- [53] A. Kostyniuk, B. Likozar, Wet torrefaction of biomass waste into value-added liquid product (5-HMF) and high quality solid fuel (hydrochar) in a nitrogen atmosphere, *Renew. Energy* 226 (2024) 120450, <https://doi.org/10.1016/j.renene.2024.120450>.

- [54] M. Grigiante, D. Antolini, An innovative experimental pilot-plant to investigate the gasification process of biomass impact of raw and torrefied pellet on the process performances, *Energy* 332 (2025) 137209, <https://doi.org/10.1016/j.energy.2025.137209>.
- [55] J. de Oliveira Brotto, D. Danaci, P.S. Fennell, J. da Silveira Salla, H.J. José, R. de Fátima Peralta, M. Moreira, Enhancing low-carbon iron and steel production with torrefied biomass, *Biomass Bioenergy* 193 (2025) 107558, <https://doi.org/10.1016/j.biombioe.2024.107558>.
- [56] R. Liao, X. Tian, L. Fan, H. Li, Y. Zeng, L. Ke, Q. Wu, R. Ruan, Y. Liu, Y. Wang, Effect of torrefaction pretreatment on catalytic pyrolysis of bagasse: a study on torrefaction method and atmosphere, *Chem. Eng. J.* 507 (2025) 160842, <https://doi.org/10.1016/j.cej.2025.160842>.